THE DERIVED CATEGORY OF QUASI-COHERENT SHEAVES IS A DERIVED CATEGORY OF A DGA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the source gives the full assembly: generator choice, finite-Ext computation, fully faithful tensor functor with adjoint, vanishing kernel on the generator subcategory and the formal equivalence criterion

Editorial note. The label theorem-DQCoh-is-Ddga names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own theorem environment in the chapter source.

Allowed standard prerequisites (editorial).

- the Bondal-Van den Bergh theorem producing a perfect K-injective generator of D_QCoh(O_X) for quasi-compact quasi-separated X
- K-injective complexes and the identification of dga cohomology with Ext groups
- the dga tensor functor - tensor^L_E K and its adjoint RHom(K, -)
- compactness of perfect objects and the fully-faithfulness criterion of the tensor functor in the compact case
- the generator property and the formal equivalence criterion for an adjunction whose restricted kernel vanishes

Source reference. Original lemma 09J1: sources/stacks-chow/tags.txt, source lines 12059–12059. Prerequisite pointer invoked at proof line(s) 4454; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09LZ: sources/stacks-chow/tags.txt, source lines 12162–12162. Prerequisite pointer invoked at proof line(s) 4436; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09LX: sources/stacks-chow/tags.txt, source lines 12160–12160. Prerequisite pointer invoked at proof line(s) 4430; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09LY: sources/stacks-chow/tags.txt, source lines 12161–12161. Prerequisite pointer invoked at proof line(s) 4441; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original section 09LU: sources/stacks-chow/tags.txt, source lines 12157–12157. Prerequisite pointer invoked at proof line(s) 4414; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09M4: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 4328–4328. Prerequisite pointer invoked at proof line(s) 4422; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09M3: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 4294–4294. Prerequisite pointer invoked at proof line(s) 4444; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09M1: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 4096–4096. Prerequisite pointer invoked at proof line(s) 4432; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09IS: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3891–3891. Prerequisite pointer invoked at proof line(s) 4406; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. Then there exist a differential graded algebra (E, d) with only a finite number of nonzero cohomology groups $H^i(E)$ such that $D_{QCoh}(\mathcal{O}_X)$ is equivalent to $D(E, d)$.*

Proof. Let $K^\bullet$ be a K-injective complex of $\mathcal{O}$-modules which is perfect and generates $D_{QCoh}(\mathcal{O}_X)$. Such a thing exists by Theorem 09IS and the existence of K-injective resolutions. We will show the theorem holds with

$$(E, \mathrm{d}) = \mathrm{Hom}_{\mathrm{Comp}^{dg}(\mathcal{O}_X)}(K^\bullet, K^\bullet)$$

where $\mathrm{Comp}^{dg}(\mathcal{O}_X)$ is the differential graded category of complexes of $\mathcal{O}$-modules. Please see Differential Graded Algebra, Section 09LU. Since $K^\bullet$ is K-injective we have

$$H^n(E) = \mathrm{Ext}^n_{D(\mathcal{O}_X)}(K^\bullet, K^\bullet) \tag{0.1.1}$$

for all $n \in \mathbf{Z}$. Only a finite number of these Exts are nonzero by Lemma 09M4. Consider the functor

$$- \otimes^{\mathbf{L}}_E K^\bullet : D(E, \mathrm{d}) \longrightarrow D(\mathcal{O}_X)$$

of Differential Graded Algebra, Lemma 09LX. Since $K^\bullet$ is perfect, it defines a compact object of $D(\mathcal{O}_X)$, see Proposition 09M1. Combined with (0.1.1) the functor above is fully faithful as follows from Differential Graded Algebra, Lemmas 09LZ. It has a right adjoint

$$R\,\mathrm{Hom}(K^\bullet, -) : D(\mathcal{O}_X) \longrightarrow D(E, \mathrm{d})$$

by Differential Graded Algebra, Lemmas 09LY which is a left quasi-inverse functor by generalities on adjoint functors. On the other hand, it follows from Lemma

09M3 that we obtain

$$- \otimes^{\mathbf{L}}_E K^\bullet : D(E, \mathrm{d}) \longrightarrow D_{QCoh}(\mathcal{O}_X)$$

and by our choice of $K^\bullet$ as a generator of $D_{QCoh}(\mathcal{O}_X)$ the kernel of the adjoint restricted to $D_{QCoh}(\mathcal{O}_X)$ is zero. A formal argument shows that we obtain the desired equivalence, see Derived Categories, Lemma 09J1. □